

CECRET: A Nextflow pipeline for reference-based amplicon consensus generation

Erin L. Young¹, Thomas Iverson¹, John Arnn¹, Olinto Linares-Perdomo², Robert Sainsbury¹, and Elizabeth Anderson¹

¹ Utah Public Health Laboratory, Department of Health and Human Services, State of Utah, Salt Lake City, Utah, USA ² Huntsman Cancer Institute, Department of Oncological Sciences, University of Utah, Salt Lake City, Utah, USA

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Summary

CECRET is an open-source bioinformatics workflow for reference-based consensus genome generation from amplicon sequencing data. Developed at the Utah Public Health Laboratory (UPHL), the workflow is optimized for Illumina sequencing using ARTIC-style primer schemes and was designed to support high-throughput viral genomic surveillance in public health laboratories (Kent et al., 2024). The pipeline automates preprocessing, alignment, primer trimming, consensus generation, quality assessment, and lineage assignment to produce standardized analysis-ready outputs from raw sequencing reads.

Unlike many early viral surveillance workflows that were primarily designed for Oxford Nanopore Technologies (ONT) data, (Quick et al., 2017) CECRET emphasizes accurate primer trimming, depth-aware filtering, and reproducible consensus generation for amplicon-based Illumina sequencing. The workflow is implemented in Nextflow with containerized dependencies, (Di Tommaso et al., 2017; K. R. Florek et al., 2025; Kurtzer et al., 2017) enabling reproducible execution across local, high-performance computing, cloud, and restricted-access environments. Although initially developed for SARS-CoV-2 surveillance, CECRET has also been adapted for pathogens including MPOX and Measles virus. The software is released under the MIT license and is publicly available through GitHub and Zenodo (Young & others, 2026).

Statement of Need

Amplicon-based sequencing protocols derived from the ARTIC framework became widely adopted during the SARS-CoV-2 pandemic because they enabled rapid and cost-effective viral genomic surveillance (Quick et al., 2017). Many public health laboratories implemented these protocols using Illumina sequencing platforms (Bull et al., 2020; Grubaugh et al., 2019). However, early bioinformatics workflows were primarily optimized for ONT data and did not consistently enforce processing steps required for reliable Illumina amplicon consensus generation, including post-alignment primer trimming and depth-aware consensus filtering.

CECRET was developed to provide a standardized, reference-based workflow specifically tailored to amplicon-based Illumina sequencing in public health surveillance settings. In these settings, accurate consensus generation requires post-alignment primer trimming and depth-aware variant filtering to avoid technical artifacts introduced during PCR amplification (Bull et al., 2020; Grubaugh et al., 2019). The workflow encodes operational best practices for preprocessing, alignment, primer trimming, consensus generation, and lineage assignment while minimizing

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41 manual parameter tuning and local customization. This design emphasizes reproducibility and
42 consistency across laboratories generating large volumes of surveillance data.

43 General-purpose workflows such as nf-core/viralrecon ([contributors, 2024](#)) support diverse
44 sequencing strategies including metagenomics and de novo assembly. In contrast, CECRET
45 prioritizes a constrained, production-oriented design optimized for high-throughput viral
46 surveillance using small reference-guided genomes. The workflow is intended for public
47 health laboratories, clinical genomics groups, and researchers requiring rapid and standardized
48 consensus genome generation from amplicon sequencing data.

49 State of the Field

50 Several workflows are available for viral genome reconstruction and genomic surveillance,
51 including the ARTIC bioinformatics pipelines ([Quick et al., 2017](#)) and more general frameworks
52 such as nf-core/viralrecon ([contributors, 2024](#)). These workflows provide broad support
53 for diverse sequencing platforms, metagenomic analyses, and flexible research-oriented
54 configurations.

55 CECRET was developed to address a narrower operational niche: high-throughput consensus
56 genome generation from amplicon-based Illumina sequencing in public health laboratory settings.
57 In contrast to general-purpose workflows that prioritize flexibility across sequencing strategies,
58 CECRET emphasizes constrained, reference-guided analysis with standardized defaults and
59 minimal configuration requirements. The workflow encodes practices commonly required
60 for Illumina amplicon surveillance, including post-alignment primer trimming, depth-aware
61 consensus generation, and integrated lineage assignment.

62 Developing CECRET as a standalone workflow rather than contributing to an existing
63 general-purpose framework allowed these operational assumptions and public health reporting
64 requirements to be implemented as core workflow behaviors rather than optional configuration
65 layers. This design prioritizes reproducibility, portability, and consistency across laboratories
66 processing large volumes of surveillance samples under routine production conditions.

67 Build vs. Contribute Justification

68 CECRET was developed as a standalone workflow because existing pipelines prioritized either
69 ONT sequencing or broad multi-platform flexibility rather than operational standardization for
70 Illumina amplicon surveillance. Public health laboratories required a workflow that enforced
71 domain-specific best practices — including post-alignment primer trimming, depth-aware
72 consensus generation, and standardized lineage assignment — as default behaviors rather than
73 optional configuration choices.

74 Implementing these assumptions within a dedicated workflow enabled a constrained, production-
75 oriented design optimized for reproducibility, minimal parameter tuning, and consistent outputs
76 across laboratories. In contrast, general-purpose frameworks such as nf-core/viralrecon
77 ([contributors, 2024](#)) are designed to support a broader range of sequencing strategies and
78 analytical objectives.

79 This design philosophy also enabled tighter integration of pathogen-specific surveillance utilities,
80 quality-control metrics, and operation in restricted or air-gapped public health computing
81 environments.

82 Software Design

83 CECRET is implemented in Nextflow DSL2 ([Di Tommaso et al., 2017](#)), enabling modular
84 workflow composition, reproducible execution, and portability across local systems, high-
85 performance computing clusters, and cloud environments. The workflow is organized into
86 subworkflows representing major stages of amplicon-based consensus generation, including

preprocessing, alignment, primer trimming, consensus generation, quality assessment, and downstream lineage analysis.

A central design goal of CECRET was operational standardization for public health surveillance workflows. Rather than maximizing configurability across sequencing strategies, the pipeline adopts a constrained reference-guided architecture optimized for Illumina amplicon sequencing of small viral genomes. This design reduces the need for local customization and enforces consistent analytical behaviors across laboratories and sequencing runs.

The workflow integrates established bioinformatics tools for alignment, primer trimming, consensus generation, and lineage assignment, while exposing only a limited set of operationally relevant parameters. Depth-aware filtering and post-alignment primer trimming are implemented as core workflow behaviors to reduce amplification artifacts and improve reproducibility of consensus genome generation from tiled amplicon sequencing data (Bull et al., 2020).

All dependencies are distributed through containerized environments using Docker, Singularity, or Apptainer (Kurtzer et al., 2017), minimizing variability caused by local system configuration. Predefined execution profiles support local, Slurm, and containerized execution environments, while continuous integration testing and parameter validation improve reproducibility and workflow stability.

Because public health laboratories frequently operate in restricted-access or air-gapped computing environments, CECRET was designed to minimize runtime dependence on external network resources. The workflow supports predownloaded reference datasets, bundled lineage assignment resources, and optional disabling of database updates to ensure stable execution in secure environments.

Optional downstream modules support phylogenetic analysis, SNP distance calculation, and wastewater lineage deconvolution, allowing the workflow to extend beyond consensus generation while preserving the same reproducible execution framework.

Research Impact Statement

CECRET has been used operationally for viral genomic surveillance in public health laboratories during the SARS-CoV-2 pandemic and subsequent outbreak responses. At the Utah Public Health Laboratory (UPHL), the workflow supported routine generation of consensus genomes and lineage assignments from Illumina amplicon sequencing data contributing to statewide surveillance efforts (BioProject PRJNA614995). The workflow has subsequently been adapted for additional pathogens including MPOX (PRJNA843095) and Measles virus (PRJNA1293457).

The software is distributed through the StaPH-B toolkit (Kelsey R. Florek & StaPH-B Contributors, 2024), supporting adoption across public health laboratory networks and integration within a broader ecosystem of reproducible bioinformatics workflows. The repository includes continuous integration testing, predefined execution profiles, and reproducible test datasets to support validation across computing environments.

Within the StaPH-B consortium, CECRET is widely used for SARS-CoV-2 consensus genome generation from Illumina amplicon sequencing data (Centers for Disease Control and Prevention, 2020). The CDC Enteric Diseases Laboratory Branch, with support from the Respiratory Viruses Branch, developed SC2CLIA Cecret, a CLIA-ready derivative workflow that extends CECRET with CDC-specific quality-control metrics, reporting, and automated NCBI submission capabilities (CDC Enteric Diseases Laboratory Branch and Respiratory Viruses Branch, 2021).

Together, these deployments demonstrate the workflow's utility as a reproducible and production-oriented platform for high-throughput viral genomic surveillance in operational public health settings.

133 Validation and Benchmarking

134 CECRET was evaluated on 21 amplicon sequencing samples (15 SARS-CoV-2 and 6 Measles
135 virus) and compared against nf-core/viralrecon. Consensus genomes generated by both
136 workflows were aligned to common reference genomes and compared using pairwise SNP
137 differences and lineage assignment concordance.

138 Across the 19 samples successfully processed by both workflows, consensus sequences differed
139 by a median of 0 SNPs (range: 0–7). CECRET successfully generated consensus genomes
140 for all 21 samples, while nf-core/viralrecon failed on two low-coverage samples (<100,000
141 reads). Lineage assignment concordance was 76.9%, with discrepancies primarily occurring in
142 low-quality or high-N consensus genomes classified as “Unassigned.” Measles clade assignment
143 showed 100% concordance across all samples.

144 Discrepancies between CECRET and nf-core/viralrecon were primarily driven by differences in
145 consensus frequency thresholds and pipeline quality gates. nf-core/viralrecon enforces strict
146 mapping and coverage thresholds that caused it to abort processing on two low-coverage
147 samples (<100,000 reads). In contrast, CECRET is designed to complete execution on low-
148 depth samples by outputting consensus genomes heavily masked with Ns at low-coverage
149 positions. Furthermore, subtle SNP differences (0–7 SNPs) occurred at sites with minor
150 intra-host variation due to differing consensus allele frequency cutoffs (0.6 in CECRET vs. 0.75
151 in nf-core/viralrecon), which subsequently influenced Pangolin lineage assignments in high-N
152 samples.

153 AI Usage Disclosure

154 Generative AI tools (ChatGPT and Google Gemini) were used to assist with debugging, workflow
155 refactoring, documentation drafting, and manuscript outlining. All generated code, text, and
156 architectural suggestions were manually reviewed and validated by the maintainer for technical
157 accuracy and biological relevance.

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